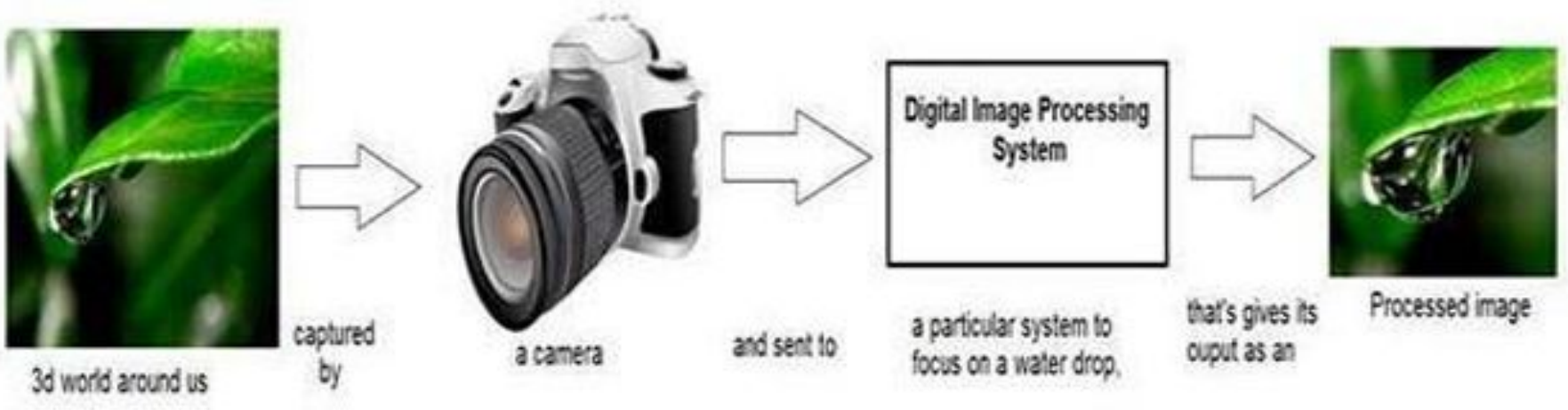
Image Processing & Computer Vision (UDSPC0603)

Fundamental of Image processing

DIGITAL IMAGE FUNDAMENTALS:

- Digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels and pixels. Pixel is the term used most widely to denote the elements of digital image.
- An image is a two-dimensional function that represents a measure of some characteristic such as brightness or color of a viewed scene. An image is a projection of a 3-D scene into a 2D projection plane
- An image may be defined as a two-dimensional function f(x,y), where x and y are spatial (plane) coordinates, and the amplitude of f at any pair of coordinates (x,y) is called the intensity of the image at that point.



- Monochrome Images: Monochrome images are digital images that contain only a single color or varying shades of a single color, typically represented as different intensity levels of gray (black to white), where each pixel carries only intensity information and no color information.
- The RGB color system, a color image consists of three (red, green and blue) individual component images.
- An image may be continuous with respect to the x- and y- coordinates and also in amplitude. Converting such an image to digital form requires that the coordinates, as well as the amplitude, be digitized.

APPLICATIONS OF DIGITAL IMAGE PROCESSING

Digital image processing has a broad spectrum of applications, such as

1. Medical Imaging

- Used for diagnosis and treatment planning in healthcare. Applied to images from X-ray, MRI, CT, Ultrasound, PET.
- Image enhancement improves contrast and visibility of tissues. Image segmentation isolates organs, tumors, and lesions.
- Pattern recognition helps in disease classification. Used for detection of tumors, fractures, and organ abnormalities. Helps in monitoring disease progression and treatment response.

2. Remote Sensing and Satellite Imaging

- Processes images captured by satellites and airborne sensors. Used for land-use and land-cover analysis.
- Helps in crop monitoring and agricultural planning. Applied in forest management and deforestation analysis. Used for weather forecasting and climate studies.
- Assists in mineral and natural resource exploration. Plays a key role in disaster management (floods, earthquakes, cyclones).Enables large-area environmental monitoring.

3. Computer Vision and Robotics

- Enables machines to interpret and understand visual data. Used in object detection and object recognition.
- Applied in face and gesture recognition systems. Supports autonomous vehicles for lane and obstacle detection.
- Helps robots in navigation and path planning. Enables human–machine interaction. Improves decision-making in intelligent systems.

4. Security and Surveillance

- Used for motion detection in surveillance videos. Tracks moving objects and suspicious activities.
- Applied in facial recognition systems. Used in biometric authentication (fingerprint, iris, face).Helps in crime detection and forensic analysis.
- Enhances public safety monitoring. Used in smart surveillance and access control systems.

5. Document Processing and Character Recognition

- Used for Optical Character Recognition (OCR).Converts scanned documents into editable text. Applied in signature verification systems.
- Used in automated form and cheque processing. Widely used in banking and postal services. Supports office automation and document digitization. Reduces manual data entry and processing time.

6. Multimedia and Entertainment

- Used for image and video enhancement. Applied in video editing and post-processing. Enables animation and special visual effects. Used in film, television, and digital media production.
- Supports gaming graphics and simulations. Plays a role in virtual reality (VR) and augmented reality (AR).Enhances overall visual experience and realism.

7. Remote Sensing and Satellite Imaging

- Processes images captured by satellites and airborne sensors. Used for land-use and land-cover analysis. Helps in crop monitoring and agricultural planning.
- Applied in forest management and deforestation analysis. Used for weather forecasting and climate studies.
- Assists in mineral and natural resource exploration. Plays a key role in disaster management (floods, earthquakes, cyclones).Enables large-area environmental monitoring.

Color Spaces

- A **color space** is a mathematical model used to represent colors in a standardized format. In digital image processing, color spaces help in efficient color representation, manipulation, enhancement, compression, and analysis.
- Different color spaces are designed to suit different applications such as visualization, segmentation, compression, and color correction.

RGB Color Space

- The **RGB (Red, Green, Blue)** color space is the most commonly used color model in digital imaging and display systems. In this model, colors are represented as combinations of red, green, and blue primary color intensities. Each pixel is defined by three values corresponding to the R, G, and B components.
- RGB is a **device-dependent color space**, meaning the same RGB values may appear differently on different devices. It is mainly used in cameras, scanners, monitors, and image acquisition systems. However, RGB is sensitive to illumination changes and is not ideal for color-based segmentation.
- **Applications:** Digital cameras, displays, image acquisition, basic image processing.

HSV Color Space

- The **HSV (Hue, Saturation, Value)** color space represents colors in a way that closely matches human color perception. **Hue** represents the type of color, **Saturation** indicates color purity, and **Value** represents brightness.
- HSV separates color information from intensity, making it more robust to lighting variations. This separation makes HSV highly suitable for color-based object detection and segmentation tasks.
- **Applications:** Object detection, color segmentation, image editing, computer vision.

- The **LAB (CIELAB)** color space is a **perceptually uniform and device-independent** color model developed by the International Commission on Illumination (CIE). It consists of three components: **L*** for lightness, **a*** representing green–red axis, and **b*** representing blue–yellow axis.

Fundamental Steps in Digital Image Processing

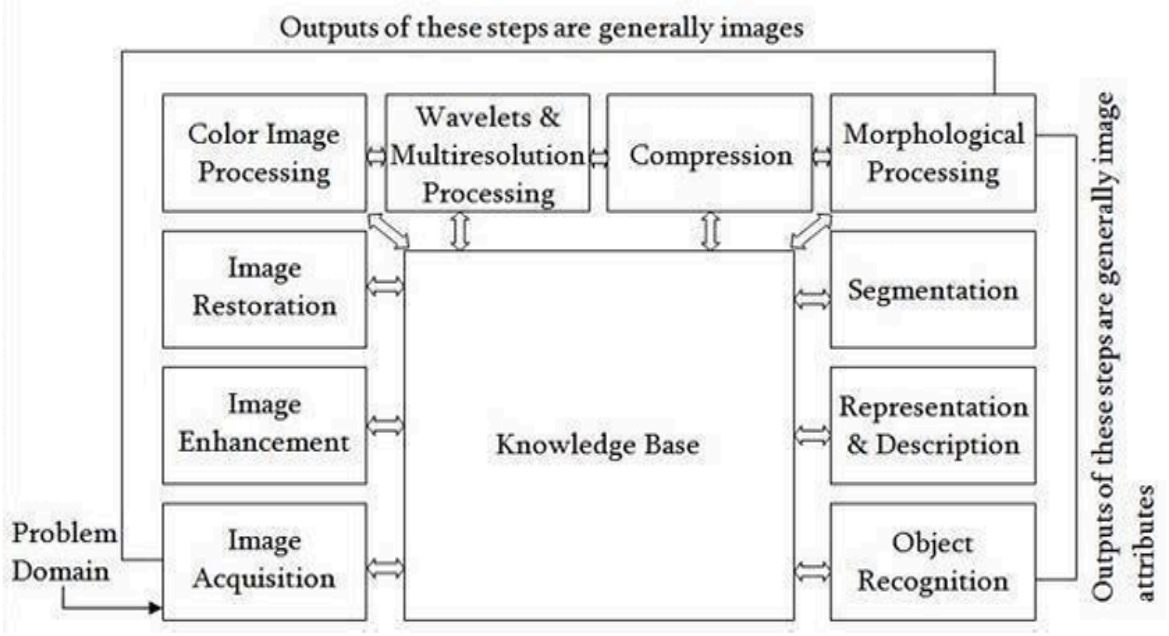


Fig: Fundamental Steps in Digital Image Processing

Digital Image Processing is a technique used to **process and analyze digital images using computer algorithms in order to extract useful information**. It plays an important role in many fields such as **medical imaging, satellite imaging, security systems, robotics, and computer vision**.

1. Image Acquisition

Image acquisition is the first and most important step in digital image processing.

In this stage, an image is captured using an image sensor such as a digital camera, scanner, or satellite sensor. The captured image is usually in analog form, so it must be converted into digital form using a process called digitization.

Digitization consists of two main processes:

• Sampling – dividing the image into small pixels
• Quantization – assigning intensity values to each pixel

After acquisition, some basic preprocessing such as noise removal or smoothing may also be performed.

Example:
Capturing an X-ray image in a hospital or capturing satellite images of earth.

2. Image Enhancement

Image enhancement is used to improve the visual appearance of an image so that it becomes easier for humans to interpret.

Sometimes the captured image may be too dark, too bright, or have poor contrast. Enhancement techniques improve these problems by adjusting brightness, contrast, or highlighting important features.

Common enhancement techniques include:

- Histogram Equalization
- Image Sharpening
- Noise Filtering
- Contrast Stretching

For example, in medical imaging, enhancement helps doctors clearly observe small structures such as tumors or fractures.

Thus, the main objective of image enhancement is to make the image more suitable for visual interpretation.

3. Image Restoration

Image restoration is the process of recovering the original image from a degraded image.

Images may become degraded due to various factors such as:

- Motion blur
- Camera misfocus
- Noise during transmission
- Atmospheric disturbances

Unlike image enhancement, restoration uses mathematical and statistical models to remove degradation and reconstruct the original image.

Example:
Removing blur from a motion blurred photograph or removing noise from a satellite image.

4. Color Image Processing

Color image processing deals with processing and analyzing images that contain color information.

Most digital images are represented using RGB color components (Red, Green, Blue). Processing color images provides more information than grayscale images and helps in better object identification.

5. Wavelets and Multiresolution Processing

Wavelet and multiresolution processing represent an image at different levels of resolution.

At low resolution, the image shows the overall structure or general appearance.
At high resolution, the image shows fine details such as edges, textures, and small objects.

This method is useful because the human visual system also observes images in multiple resolutions.

Applications include:

• Image compression
• Noise removal
• Image analysis

Example:
When we zoom an image on a smartphone, we first see a blurred version, and gradually the details become clearer.

6 Compression

Image compression is used to reduce the size of an image file.

Explanation

It removes redundant data so that the image requires less storage space and less transmission bandwidth.

Types

1. Lossless Compression – No data loss (PNG, TIFF)
2. Lossy Compression – Some data loss but higher compression (JPEG)

7. Morphological Processing

Morphological processing is used to analyze the shape and structure of objects in an image.

Explanation

It performs operations based on the geometrical structure of objects.

Common operations

Dilation, Erosion, Opening, Closing

Applications

- Shape analysis
- Object boundary detection

8. Image Segmentation

Image segmentation is the process of dividing an image into several meaningful regions.

The goal of segmentation is to separate objects from the background so that they can be analyzed individually.

Segmentation is usually based on properties such as:

Pixel intensity, Color, Texture, Edges

Example:
In medical imaging, segmentation can separate tumor regions from normal tissues.

Segmentation is one of the most important steps in image analysis.

9. Object Recognition

Object recognition is the final stage of digital image processing.

Explanation

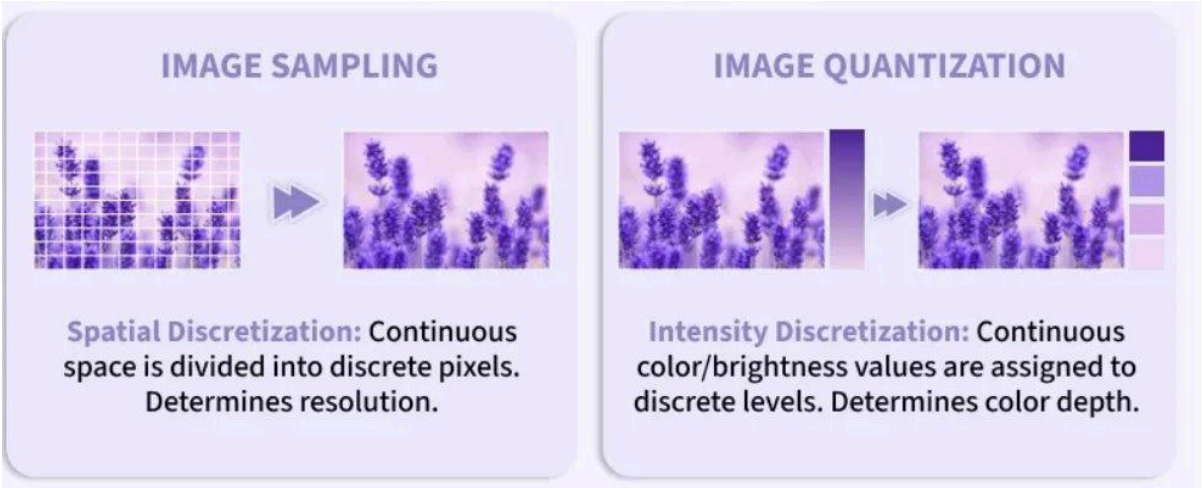
In this step, the system identifies and labels the objects present in the image based on extracted features.

Example

Recognizing objects such as:

- Car
- Person
- Tumor in medical images

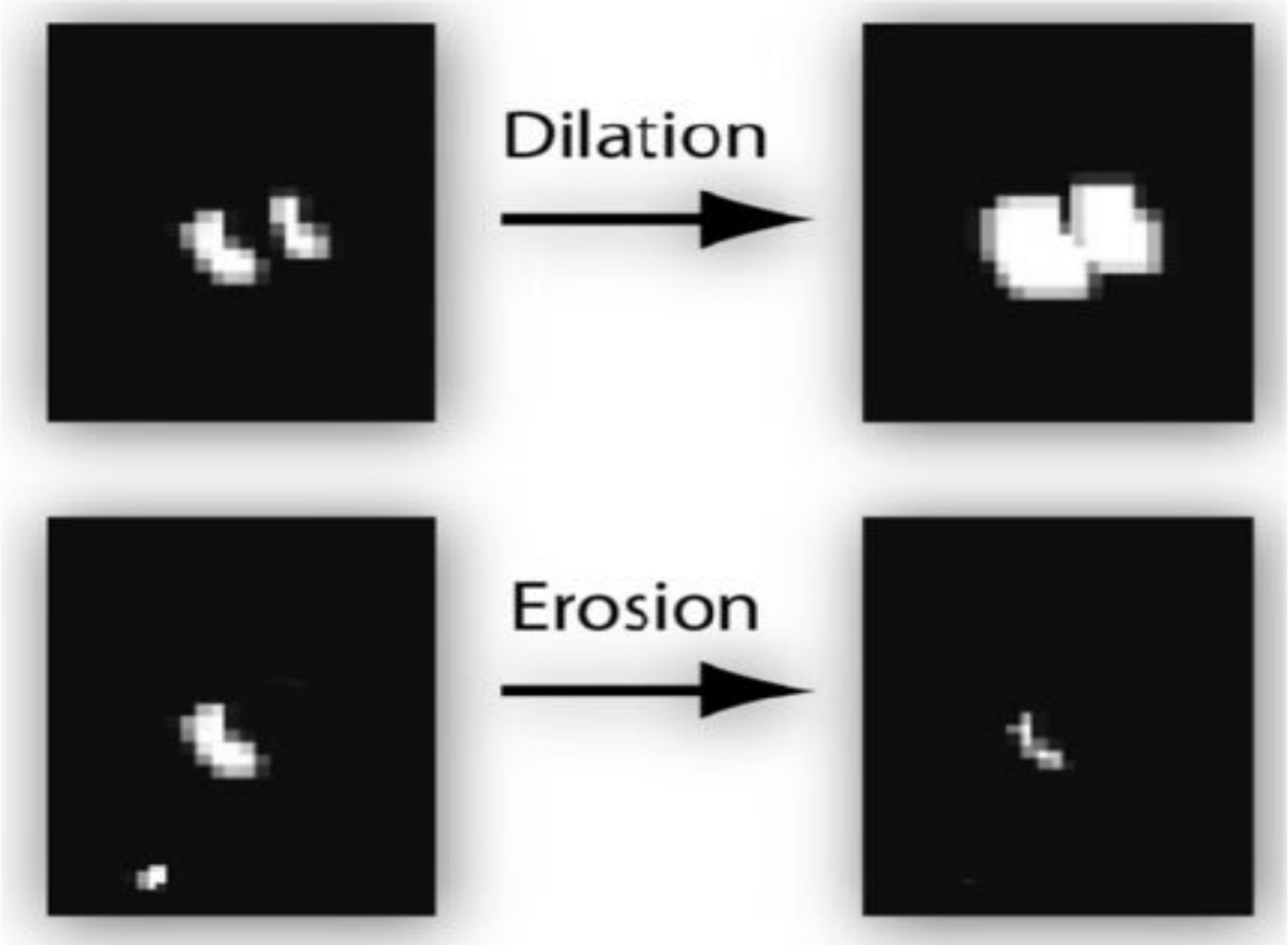
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Fundamentals of Morphological Operations in Image Processing

Morphological Operations

- Morphological operations rely on two key elements: the input image and the structuring element. The input image is typically a binary image in which the objects of interest are represented by foreground pixels and the background by background pixels. Grayscale images can also be processed to enhance or suppress specific intensity patterns.
- The structuring element is a small matrix or kernel that defines the neighborhood of pixels examined during the operation. The shape, size, and orientation of the structuring element significantly influence the outcome, making it a critical design parameter in morphological analysis.


Dilation
Erosion

1. Erosion

- Erosion is a basic morphological operation that **shrinks object boundaries** by removing pixels from the edges of foreground regions. It is commonly used to **eliminate small noise**, **separate connected objects**, and **thin object structures**.
- During erosion, only those pixels that fully accommodate the structuring element remain in the output, causing objects to gradually reduce in size and lose fine details.

2. Dilation

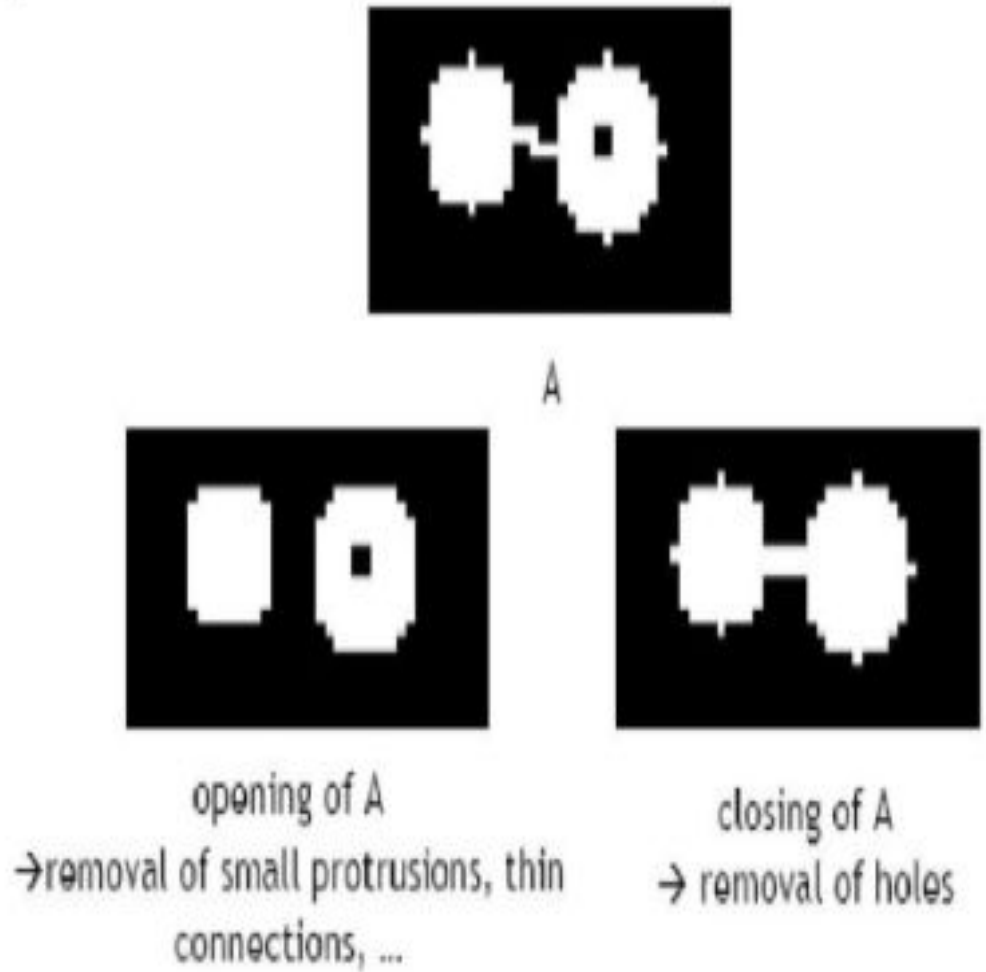
- Dilation is the complementary operation to erosion and is used to **expand the boundaries of objects** in an image. It is effective for **filling small holes**, **bridging narrow gaps**, and **connecting nearby objects**.
- In dilation, the presence of any foreground pixel within the neighborhood defined by the structuring element results in the expansion of object regions, making features appear larger and more prominent.

3. Opening

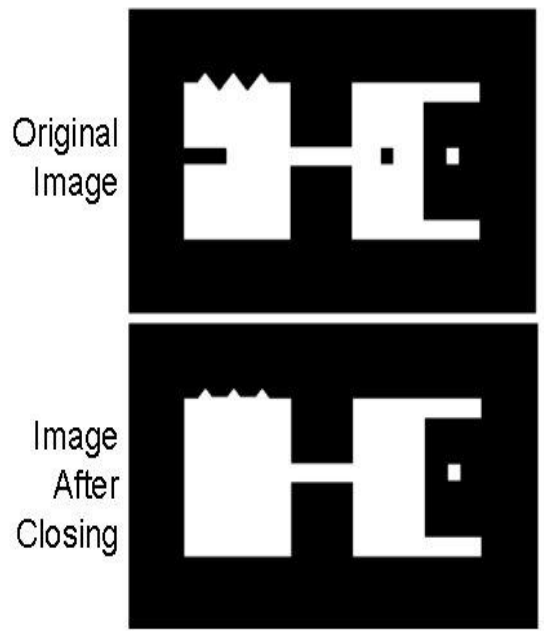
- Opening is a **compound morphological operation** that performs erosion followed by dilation. It is mainly used to **remove small unwanted objects or noise** while preserving the **overall shape of larger structures**.
- Opening smoothens object contours, breaks thin connections, and eliminates minor protrusions without significantly affecting the main regions in the image.

4. Closing

- Closing is another compound operation that consists of dilation followed by erosion. It is particularly useful for **filling small holes and gaps** within objects and **closing breaks in boundaries**.
- Closing helps in producing smoother and more continuous object regions while maintaining their general size and shape.
- Open : Erosion then dilation
- Close : Dilation then Erosion



Closing: Example

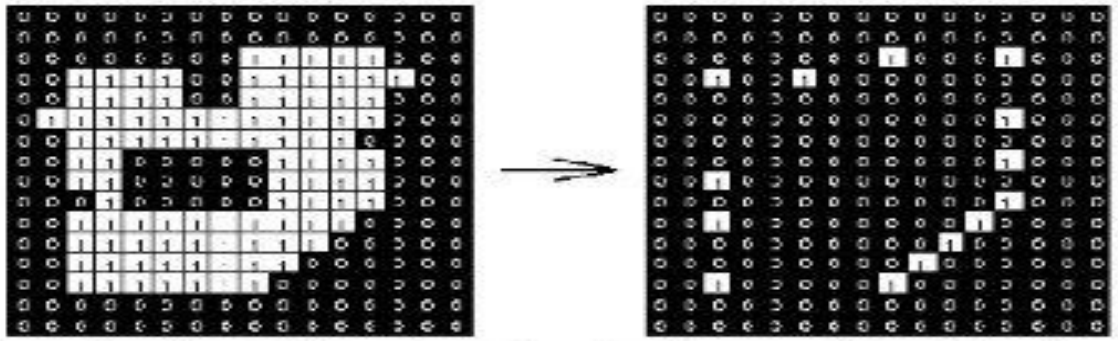


Closing
Eliminates small holes
Fills gaps
Removes 'Pepper' noise

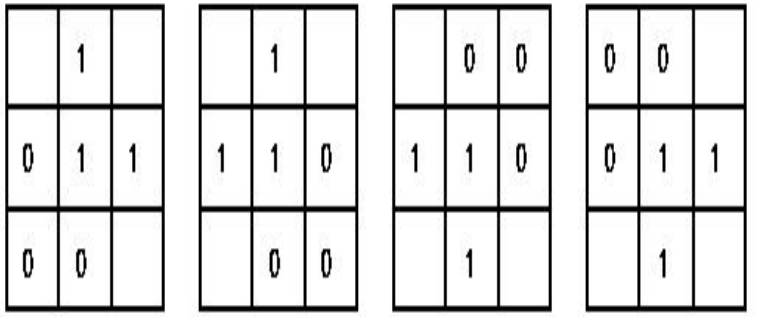
5. Hit-or-Miss Transform

- The hit-or-miss transform is a morphological technique designed to **detect specific patterns or configurations** in a binary image. It uses two structuring elements—one to match the foreground and another to match the background.
- This operation is especially useful in **shape detection**, **template matching**, and **structural analysis** applications.

1. Apply Hit-&-Miss with each Structuring Element
2. Use OR operation to combine the four results



Allows to describe approximate shape of an object



Foreground Pixel

A **foreground pixel** represents the **object or region of interest** in an image.

- In a **binary image**, it is usually represented by **value 1** or **white color**.
- Foreground pixels form the **shape, structure, and boundaries** of objects.
- In morphological operations, foreground pixels are the **primary focus** for analysis such as erosion, dilation, and hit-and-miss transform.

Example:
In a binary image of text, the letters are made up of foreground pixels.

Background Pixel

A **background pixel** represents the **area outside the object of interest**.

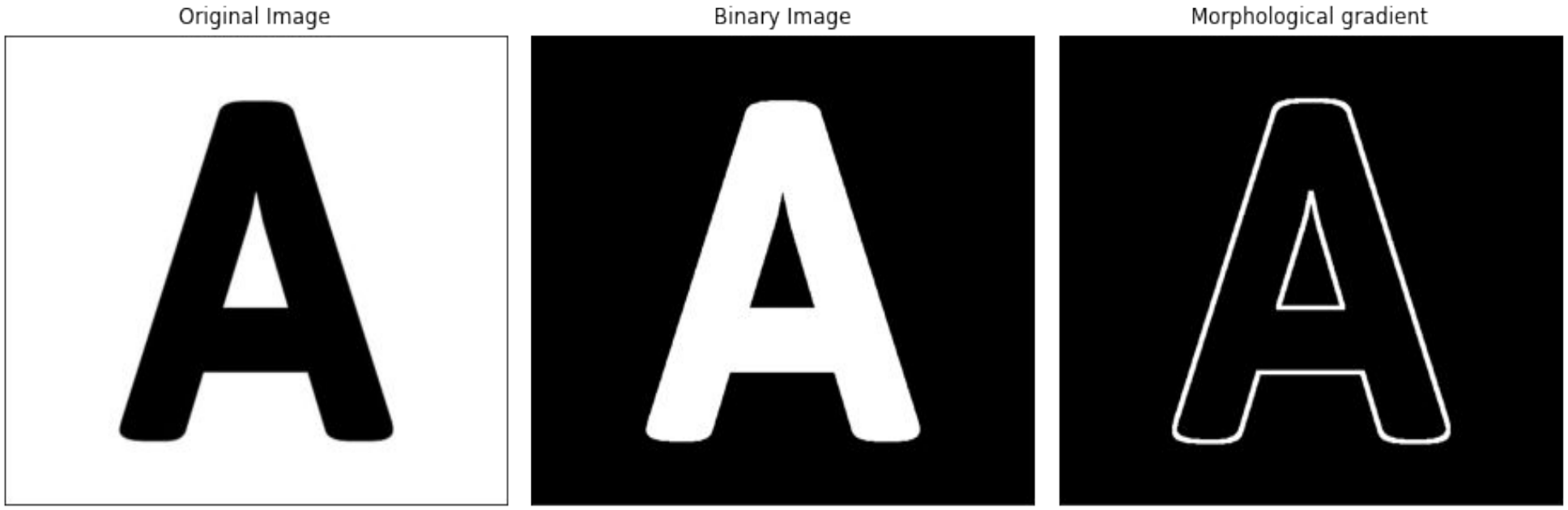
- In a **binary image**, it is usually represented by **value 0** or **black color**.
- Background pixels provide contrast and help in identifying object boundaries.
- In morphological operations, background pixels are important for detecting gaps, holes, and object edges.

Example:
In a binary image of text, the white area around the letters is the background.

- The **top-most 3×3 blocks are morphological filters**, more precisely called **structuring elements** used in the **hit-and-miss transform**.
- Each **3×3 structuring element** represents a **right-angle convex corner** in a specific orientation.
- These structuring elements are **slid pixel by pixel** over the bottom-left binary image (just like convolution, but logical matching instead of multiplication).
- At every position:
- Pixels marked **1** in the filter must match **foreground pixels** in the image.
- Pixels marked **0** in the filter must match **background pixels**.
- Blank positions are ignored (“don’t care”).

6. Morphological Gradient

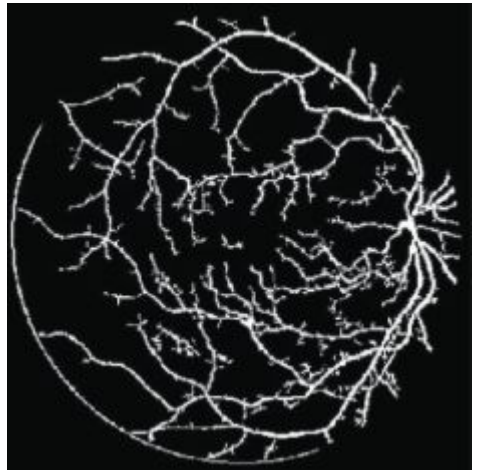
- The morphological gradient is used to **highlight the boundariesor edges** of objects in an image.
- By emphasizing the transition between foreground and background regions, this operation helps in **edge detection and boundary extraction**, making it useful for segmentation and shape analysis tasks.



7. Top-Hat Transform

- The top-hat transform is applied to **extract small details and fine features** from an image. It is commonly used to **enhance bright objects on a dark background** or **dark objects on a bright background**.
- This transform is especially effective in correcting uneven illumination and enhancing subtle features that may otherwise be overlooked.





Diagram

```
(x-1,y+1)   (x,y+1)   (x+1,y+1)

(x-1,y)     P(x,y)    (x+1,y)

(x-1,y-1)   (x,y-1)   (x+1,y-1)
```

Consider a Pixel

Let the given pixel be **P(x, y)**

Its neighbours are defined based on their position relative to this pixel.

🔹 Types of Pixel Neighbours

1. 4-Neighbours (N_4)

These include pixels that share a **side (horizontal or vertical)** with the central pixel.

Coordinates:

- (x+1, y)
- (x-1, y)
- (x, y+1)
- (x, y-1)

📌 So,
$N_4(P)$ = {(x+1,y), (x-1,y), (x,y+1), (x,y-1)}

---

2. Diagonal Neighbours (N_D)

These include pixels that are placed **diagonally**.

- (x+1, y+1)
- (x+1, y-1)
- (x-1, y+1)
- (x-1, y-1)

---

3. 8-Neighbours (N_8)

This is the combination of **4-neighbours + diagonal neighbours**.

So,
$N_8(P) = N_4(P) + N_D(P)$

Total = 8 neighbours

Basic relationships between pixels (Adjacency):

- Connectivity between pixels is a fundamental concept that simplifies the definition of numerous digital image concepts, such as regions and boundaries.

(a) 4 - adjacency: Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.

(b) 8 - adjacency: Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

(c) m – adjacency (mixed adjacency): Two pixels p and q with values from V are m-adjacent if (i) q is in the set $N_4(p)$, or (ii) q is in $N_D(p)$ and the set $N_4(p) \cap N_D(p)$ has no pixels whose values are from V.

V = {1}

0	1	0	1
0	0	1	0
0	0	1	0
1	0	0	0

V = {0,1,2,3,.....10}

66	9	90	7
92	166	6	19
110	66	5	77
8	80	133	70



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Types of Geometric Transformations

1. Translation

It shifts the image from one position to another.

$$x' = x + t_x, y' = y + t_y$$

Homogeneous form:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2. Rotation

It rotates the image by an angle θ about origin.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Example: Rotation by 45°

$$\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

So matrix becomes:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

This rotates the image **anticlockwise by 45°**.

3. Scaling

Used to change the size of the image.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- $S_x > 1$→ Zoom in
- $S_x < 1$→ Zoom out

4. Reflection

Used to flip the image.

- About X-axis:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- About Y-axis:

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

5. Shearing

Used to distort the image along one axis.

- X-direction:

$$\begin{bmatrix} 1 & Sh_x \\ 0 & 1 \end{bmatrix}$$

- Y-direction:

$$\begin{bmatrix} 1 & 0 \\ Sh_y & 1 \end{bmatrix}$$

```
import numpy as np
import cv2 as cv
import tensorflow as tf
import matplotlib.pyplot as plt
```

```
img = cv.imread('/content/drive/MyDrive/Colab Notebooks/Tiger.webp')
```

```
plt.imshow(img)
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1599.5), np.float64(899.5), np.float64(-0.5))
```



```
tiger = cv.cvtColor(img, cv.COLOR_BGR2RGB)
```

```
plt.imshow(tiger)
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1599.5), np.float64(899.5), np.float64(-0.5))
```



```
img_tf = tf.convert_to_tensor(tiger)

# Flip image left-right using TensorFlow
img_flipped = tf.image.flip_left_right(img_tf)

# Display result
plt.imshow(img_flipped)
plt.axis('off')
plt.show()
```



```
img_flipped = tf.image.flip_up_down(img_tf)

# Display result
plt.imshow(img_flipped)
plt.axis('off')
plt.show()
```



```
cropped_img = tiger[100:500, 100:5000]
plt.imshow(cropped_img)
plt.axis('off')
plt.show()
```



```
rotated_img = cv.rotate(tiger, cv.ROTATE_90_CLOCKWISE)
plt.imshow(rotated_img)
plt.axis('off')
plt.show()
```



```
(h, w) = tiger.shape[:2]
center = (w // 2, h // 2)
# Create rotation matrix (45 degrees clockwise  use -45)
M3 = cv.getRotationMatrix2D(center, -45, 1.0)
rotated_img = cv.warpAffine(tiger, M3, (w, h))

plt.imshow(rotated_img)
plt.axis('off')
plt.show()
```



```
M4 = cv.getRotationMatrix2D(center, 30, 1.0)

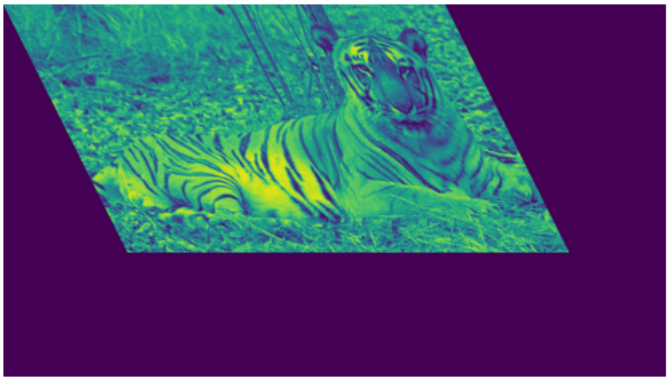
rotated_img2 = cv.warpAffine(tiger, M4, (w, h))

plt.imshow(rotated_img2)
plt.axis('off')
plt.show()
```



```
img2 = cv.imread('/content/drive/MyDrive/Colab Notebooks/Tiger.webp',0)
rows, cols = img2.shape
M = np.float32([[1, 0.5, 0], [0, 1, 0], [0, 0, 1]])
sheared_img = cv.warpPerspective(img2, M, (int(cols*1.5), int(rows*1.5)))
```

```
plt.imshow(sheared_img)
plt.axis('off')
plt.show()
```



```
M2 = np.float32([[1,   0, 0], [0.5, 1, 0], [0,   0, 1]])
sheared_img2 = cv.warpPerspective(img, M2, (int(cols*1.5), int(rows*1.5)))
```

```
plt.imshow(sheared_img2)
plt.axis('off')
plt.show()
```



```
Start coding or generate with AI.
```

```
def contrast_stretching(image):
  img_float = image.astype(np.float32)
  min_val = np.min(img_float)
  max_val = np.max(img_float)
  stretched = (img_float - min_val) * (255.0 / (max_val - min_val))
  return np.clip(stretched, 0, 255).astype(np.uint8)
```

```
stretched_image = contrast_stretching(tiger)
```

```
plt.imshow(stretched_image)
plt.axis('off')
plt.show()
```

